The Trap of Model-Derived Yield Targets: A Controlled Diagnostic of Target Construction, Validation Design, and Co-Construction Bias in County-Scale Yield Mapping

by

Yucheng Peng 1,*, [Co-author 2] 1, [Co-author 3] 1 and [Co-author 4] 1

1 College of Agronomy / National Maize Improvement Center, China Agricultural University, Beijing 100193, China

* Correspondence: [advisor email].

Abstract: Regional yield mapping relies on model-derived targets because grid-scale field measurements are rare. We audited what such targets do to the apparent skill of environmental machine learning in Pinggu, Beijing (234 grids, 0.0417 degrees, 24 variables), where a provenance audit showed that all four obtainable yield targets are model products: a process-model baseline (43 grids, 7.21 plus or minus 0.23 t/ha), its interpolation, a Bayesian blend, and a satellite retrieval. The interpolated target yields cross-validated R2 = 0.98 although leave-one-out interpolation of its own sources reaches only 0.20 to 0.27, bounding circular validation. Checkerboard cross-validation is structurally inert in a domain 5.6 variogram ranges wide (between-block distance pinned at 2.3 to 2.4 km; optimism gap from +0.113 to -0.006), whereas gradient-cutting schemes reveal gaps up to +0.54. The same framework recovers R2 = 0.690 on a 7,784-grid regional gradient yet detects nothing at n = 43 even for planted true signals of R2 = 0.5: county-scale nulls are detectability limits. Two national products correlate at r = 0.017 while one environmental model explains both at R2 near 0.65; provincial statistics arbitrate (rho = 0.893 vs -0.04), proving co-construction bias. We release a six-step pre-mapping audit protocol that turns each trap into a checkable pre-condition.

Keywords: yield mapping; spatial cross-validation; circular validation; co-construction bias; model-derived targets; detectability; audit protocol

1. Introduction

Maps of crop yield are the operational currency of food-security monitoring, crop-insurance design, and climate-impact assessment [1-3]. Because combine-measured yields are almost never available at the grid scales these applications demand, the standard workflow trains machine-learning models of yield against model-derived targets: process-based crop-model outputs [4], satellite retrievals [5], or interpolations of sparse ground points. The community has grown adept at reporting high cross-validated skill for such models; it has been far less diligent in asking what, exactly, that skill measures.

Three methodological risks travel with this workflow, each documented in isolation. First, target construction can manufacture skill: a model evaluated against a spatial interpolation of its own training inputs recovers the interpolation, not the field (circular validation) [6,7]. Second, validation design can conceal optimism: random cross-validation under spatial autocorrelation inflates apparent accuracy, and block designs are the recommended remedy [8-10]-yet the remedy itself is rarely audited for whether the blocks actually separate training from validation neighbourhoods. Third, environmental representation sets a detectability ceiling: static covariates compress an environment that crops experience dynamically, and at small spatial extents the remaining signal may fall below any honest estimator's detection floor [11,12]. What no study has done is quantify the three risks jointly, in one controlled design, on public data-and then ask the question that emerges only when they are seen together: if a model reports high skill against two targets that disagree with each other, what is it actually predicting?

We answer these questions with a diagnostic study built on Beijing's Pinggu district and the public data ecosystem around it. The study began as a conventional exercise-predict winter-wheat yield from a 234-grid environmental fingerprint-and was progressively converted into an audit when its own results failed conventional expectations. A provenance audit established that every locally available yield target, including one initially labelled "observed", is a model product (Section 2). Controlled factorial experiments then quantified each trap: circular validation bounded analytically and empirically; checkerboard blocking shown to be structurally inert in small domains through a dose-response and nearest-neighbour analysis; and the one-axis structure of the environmental matrix exposed by principal-component screening. Two external checks generalise the diagnosis: the same framework recovers strong, genuine environmental signal across a 7,784-grid North China Plain gradient (R^2=0.69), demonstrating that the county-scale null is a detectability outcome; and a statistical arbitration against official provincial surveys shows that two national yield products correlate at r=0.017 grid-to-grid while environmental models "explain" both

equally-co-construction bias, proved directionally by the fact that only one of the two products carries statistical signal.

The contribution is fourfold: (i) quantified upper bounds for the target-construction and validation-design traps; (ii) a detectability framework (domain size × environmental variance × sample size × model complexity) that reconciles "environment matters" at regional gradients with honest nulls at county scale; (iii) directional proof of co-construction bias and its antidote-inter-target agreement audit with statistical arbitration; and (iv) a six-step pre-mapping audit protocol that converts every finding into a checkable pre-condition for any regional yield-mapping study.

2. Study Area, Targets, and the Provenance Audit

2.1 Study area and environmental matrix

Pinggu district (116.90-117.26 °E, 40.00-40.24 °N) spans a mountain-to-plain transition on the northern North China Plain, with elevations from 15 to 898 m across only ~26 km. We discretised the district into 234 grids on a 0.0417° lattice (~3.6 × 4.6 km, anisotropic) and attached 24 modelling variables from WorldClim 2.1 [13] (19 bioclimatic plus monthly temperature and precipitation aggregates), SoilGrids 2.0 topsoil properties [14], and SRTM terrain derivatives. An empirical variogram of the yield-trap targets gives a spatial range of 4.62 km; the district spans only ~5.6 such ranges-a fact that becomes central in Section 5.

2.2 The four local targets and their provenance

The audit that anchors this paper established that no grid-scale field-measured yield is publicly available for the study area, and that all four obtainable targets are model products. Target A (43 grids; 7.21 ± 0.23 t/ha) is the baseline-period output of the process-model optimisation of Xiao et al. [4], extracted from their published 1-km raster (figshare DOI 10.6084/m9.figshare.24471919); its label as "observed" in early project documents was a provenance error, corrected here. Its magnitude-38% above the Beijing statistical mean of 5.24 t/ha-matches potential-yield semantics. Target B is a Ridge interpolation of A across all 234 grids (training-fit $R^2 = 0.99$). Target C blends B with a district-level prior of 6.0 t/ha. Target D is the ChinaWheatYield30m satellite retrieval for 2021 [5] (22 grids at a >=4-valid-pixel threshold; 5.82 ± 0.21 t/ha on the common extraction window), +10% above the statistical mean.

2.3 External data for generalisation and arbitration

Three public assets extend the diagnostic beyond Pinggu. (i) The full History period of the Xiao et al. panel [4], aggregated to the same lattice, yields 7,784 national grids with matching environment features; its scenario periods (2030s, 2060s under SSP2-4.5 and SSP5-8.5) provide climate-transfer testbeds, and its six management-reference columns (N rates, irrigation, and their ratios for both rotation crops) provide the management-information arm. (ii) The ChinaWheatYield30m product for 2016, 2020, and 2021 (1-km national rasters) provides the inter-annual arm. (iii) The National Bureau of Statistics 2021 summer-grain announcement provides provincial survey yields (real-cut, real-measure methodology) as an independent arbiter. All source files, checksums, and acquisition paths are released in the data manifest; large rasters ship via the repository's data release.

3. Methods

3.1 Factorial design

All experiments follow one controlled design: {feature set} × {validation scheme} × {model} × {target}, with feature sets spanning environment-only (E, 24-71 columns depending on line), management-only (M, six columns), and their union (E+M). Every cell reports pooled out-of-fold R^2 over 3-5 random seeds (mean ± SD, minimum reported), with fold-mean statistics as a secondary, explicitly labelled convention. Hyperparameters are fixed at the XGBoost configuration transferred from an independent 250-m pipeline (428 trees, depth 5, learning rate 0.168; provenance in e6_tuned_params.json); a nested re-tuning sensitivity check on the diagnostic data reproduces all qualitative conclusions (random 0.401 / checker 0.373 / lat-block 0.196).

3.2 Validation-design family and dose-response

Five schemes span the interpolation-extrapolation spectrum: random 5-fold; checkerboard blocking at 0.5×-4× the variogram range (dose-response); latitude quartile blocks; leave-one-block-out (K-means, k=8); and environment-space K-means blocks. For each scheme we compute the optimism gap (random minus scheme pooled R^2, positive = scheme more honest) and the mean nearest-neighbour distance between blocks-the mechanism probe that distinguishes genuine blocking from silent degeneration into random splits. Variogram ranges are estimated from the target field itself (Pinggu range 4.62 km; national History-yield range ~462 km), a distinction that proves decisive.

3.3 Representation diagnostics and power analysis

Principal-component screening quantifies the effective dimensionality of the "environmental fingerprint" and its elevation-axis content. Detection power is measured by planting synthetic signals of known population R^2 (0.05-0.5) into the real 43-grid feature space, using two signal shapes: adversarial random directions and the regionally learned response (the national E-model's prediction on the Pinggu grids, correlation ~0.7 with April temperature and precipitation-physically well-formed). Detection is declared at out-of-fold $R^2 > 0.10$. A weakly regularised comparator (100 trees, depth 2) tests the model-complexity-matching principle.

3.4 Uncertainty quantification

Quantile random forests (300 trees) supply 10/50/90 percentiles; split-conformal calibration (70/70/94 split, q-hat = 2.08) recalibrates the 90% intervals. Reporting follows the decision-suitability matrix: interval width is compared against the precision each use case (insurance pricing, trend screening, warning zonation, carbon MRV) actually requires.

3.5 Inter-target agreement and statistical arbitration

Wherever two independently constructed targets overlap, we report grid-level Pearson and Spearman correlations; where an official statistical anchor exists, we assign lattice cells to provinces by nearest-provincial-centroid Voronoi and compare provincial means by Spearman rank correlation against the survey. The alarm pattern motivating protocol step (vii) is low inter-target correlation combined with high per-target environmental R^2.

3.6 Reproducibility

All folds (five released fold-assignment files), seeds, parameters, and requirements are frozen in the repository; every number in this manuscript is regenerated by one of the scripted ledgers (w14-w26 series), and Section 10 lists the one-command reproduction chain.

4. Results I: The Target-Construction Trap

Against target A-the process-model baseline whose provenance is documented-the environmental models show no skill under any scheme (all pooled $R^2 <= 0.20$; random CV -0.05 ± 0.02 with tuned trees). Against target B, the Ridge interpolation of A, the same models report pooled $R^2 = 0.98$. The gap is the trap, and it is bounded: leave-one-out inverse-distance interpolation of B's own 43 source values attains only $R^2 = 0.20$-0.27, so any environmental model "predicting" B at 0.98 must be reproducing the interpolation rule itself. Two magnitude checks triangulate semantics: A sits +38% above the Beijing statistical mean (potential-yield semantics), while the satellite target D sits +10% above it (actual-yield semantics); the ~28-point gap is itself a management-gap signature (Section 7). The blended target C inherits smoothness rather than signal-its correlation with D on common grids is -0.01, the district prior having flattened what spatial structure remained.

5. Results II: The Validation-Design Trap

The five-scheme spectrum on target C reads: random 0.400 ± 0.015; checkerboard (1× range) 0.367; latitude blocks 0.113 pooled (fold-level -0.205-the two statistics disagree exactly as small-block variance theory predicts); LOBO 0.124; environment-space blocks -0.136. The dose-response experiment explains why the checkerboard family never joins this ranking: from 0.5× to 4× the variogram range, the optimism gap decays from +0.113 to -0.006 while the mean nearest-neighbour distance between blocks never leaves 2.3-2.4 km-blocks large enough to span the domain silently degenerate into random splits. In a domain only ~5.6 ranges wide, checkerboard blocking is structurally a placebo; the honest schemes are those that cut the dominant gradient (latitude +0.288, LOBO +0.276, environment-space +0.54).

The national lattice reproduces the mechanism with the sign structure intact and adds a scale correction: with the range estimated from the national yield field itself (~462 km), the North China Plain spans only ~2 of its own ranges, and checkerboard gaps never exceed +0.07 at any sub-range block size while latitude blocking costs +0.32-gradient cutting, not block size, generates informative gaps in both domains, and block sizes must be expressed in target-field range units (protocol step ii).

6. Results III: Representation and Detectability

6.1 One axis wearing twenty-four hats

The "24-variable environmental fingerprint" is, to first order, a single elevation axis: PC1 explains 66.3% of variance and correlates with elevation at r = -0.94; the top SHAP variables (bio9, December mean temperature, bio11) are

winter-temperature expressions of the same mountain-plain gradient, physically coherent with overwintering stress but rank-unstable within the collinear block. PC1-PC5 jointly explain only $R^2 = 0.24$ of target A-collinearity cannot rescue the null. Winter-wheat-appropriate thermal features (Tbase = 0 °C seasonal GDD, vernalisation days, deep-freeze days, grain-fill heat) add +0.006 on target C and exactly zero on A (bit-identical out-of-fold predictions; the four features receive zero split gain)-correcting the physiology of the covariates does not change the county-scale verdict.

6.2 The null is a detectability outcome, not a verdict on environmental control

The identical framework, features, and hyperparameters recover $R^2 = 0.690 \pm 0.006$ on the 7,784-grid national gradient (0.689 on the 25-column subset shared with Pinggu; 0.450 at the 1-km fine level, separating genuine signal from aggregation smoothing). Two experiments then locate the county-scale boundary. Sliding 60 Pinggu-sized windows across the national dataset yields a median R^2 of 0.163 (IQR 0.027-0.272; 60% of windows below 0.2): near-nulls are the norm at this domain size, not a Pinggu anomaly. And the power analysis plants regionally learned signals (true R^2pop = 0.05-0.5) into the real 43-grid feature space: tuned tree ensembles detect nothing (pooled detection probability 0.00 even at R^2pop = 0.5, predictions collapsing toward a constant, SD 0.018 against a target SD of 0.224), while a weakly regularised comparator senses the signal fold-wise (40% detection at R^2pop = 0.3) but loses it after pooling (3%). The county null is therefore the honest read of a sampling geometry whose detection ceiling lies below the true signal-model complexity must match sample size, and no re-tuning conjures skill from n = 43.

7. Results IV: Management Information and Climate Transfer

7.1 Management dominates where it can be seen

On the national lattice, six management-reference columns out-predict twenty-five environmental ones: M-only R^2 = 0.913, E-only 0.690, E+M 0.948 (random CV; checkerboard 0.897/0.625/0.932). SHAP attributes 0.95 of the M-model importance to the rotation-maize N rate-coherent with an annual-rotation target. Because the management surface and the panel yields share a process-model lineage, this number is a within-family consistency bound rather than independent field evidence (the same-source caveat that governs all Xiao-panel results); the National Bureau of Statistics arbitration in Section 8 is what separates the two data families. At Pinggu, tuned trees collapse all feature sets to constants (Section 6.2), but the weakly regularised comparator recovers management signal on the 43-grid target A (0.218 ± 0.054 across three seeds, all positive) while environment stays null (-0.17) and E+M is diluted by noise columns (0.05)-management information is detectable at county scale where environment is not, with the strict semantics caveat that target A itself is a process-model product.

7.2 Climate transfer: pattern retained, level lost

A model trained on the History period and applied unchanged to the scenario periods loses pooled R^2 from 0.96 (in-sample era) through 0.40 (2030s) to -0.95 (2060s), but spatial-pattern skill-Pearson correlation with the period's yield field-declines only from 0.961 through 0.888 to 0.861, with deeper losses under SSP5-8.5 (0.089) than SSP2-4.5. The negative pooled values are mean-shift artifacts: the static-covariate model preserves the relative map while remaining blind to the -1.4 t/ha climate-driven level change. Mapping transfers; impact assessment does not.

8. Results V: Co-Construction Bias and Statistical Arbitration

The two national products disagree almost everywhere: grid-level correlation between the satellite retrieval and the optimised panel on 5,652 common cells is r = 0.017. Yet the environmental model "explains" both at $R^2 = 0.651$ and 0.658-high skill on mutually uncorrelated targets cannot reflect a shared yield process; it reflects each product's construction rule being environment-dependent. We call this co-construction bias. Three further facts close the case. The bias is year-robust (E->RS = 0.861/0.709/0.649 for 2016/2020/2021). Inter-annual pattern agreement of the satellite product itself is intermittent-2016 vs 2021 grids correlate at 0.881 and 2017 joins the family (0.44 with both), but 2020 breaks against all three (-0.61 to -0.07, its spatial SD halved), so cross-year audits must report pairwise. And the official statistics arbitrate: provincial survey ranks match the satellite product at Spearman rho = 0.893 ($p = 0.007$; absolute errors < 0.25 t/ha) but not the optimised panel (rho = -0.036). One product carries real signal, the other does not, and the environmental model cannot tell them apart-directional proof that at least the panel share of the environmental R^2 is construction, not yield.

9. Uncertainty and Decision Value

Raw QRF 90% intervals under-cover (PICP = 0.705); split-conformal calibration lifts coverage to 0.862 at the price of width (mean prediction interval width 0.62 -> 0.79 t/ha ~ 2.6 × the target SD). The decision-suitability matrix converts

these numbers into usable statements: at current precision the intervals support trend screening and warning zonation, marginally support relative comparison, and cannot support insurance-grade pricing or carbon-MRV accounting, both of which require widths below the process noise they are meant to price. A number's value is defined by the decision it can support-an uncertainty statement without its decision context is incomplete reporting.

10. Discussion

The three traps share one mechanism. Target construction, validation design, and co-construction bias are not three failure modes but one: apparent skill accrues to whatever process generated the target's spatial structure-interpolation rules, panel optimisation logic, satellite-retrieval calibration-and honest models inherit that structure unless the evaluation design separates them. Our controlled factorials make each inheritance visible and, importantly, measurable in advance: the circular upper bound from a leave-one-out interpolation, the placebo checkerboard from a nearest-neighbour distance, the co-construction alarm from an inter-target correlation.

Detectability reframes "environment does not explain yield". The same framework that yields honest nulls at Pinggu recovers $R^2 = 0.69$ across the regional gradient; the difference is geometry-environmental variance, domain size in target-field range units, and sample size-not doctrine. This is the quantitative reconciliation between the agricultural-systems consensus (environment shapes yields) and the county-scale null: the question is never whether environment matters but whether a given sampling geometry can see it. The management result sharpens the picture: where information exists (six management columns) and the estimator is matched to n (weak regularisation), county-scale detection succeeds.

Limitations. Grid-scale ground truth remains absent for the study area; our strongest arbitrated comparison lives at provincial resolution. The management arm shares lineage with its panel target (within-family evidence), and the climate-transfer arm tests transfer within one model family's scenario space. Inter-annual conclusions rest on four years at Pinggu (2016/2017/2020/2021: the three non-2020 years correlate at 0.44-0.88 while 2020 correlates negatively with all three, $r = -0.61$ to -0.07; county-scale environment-CV negative in every year, -0.14 to -0.28), with 2018-2019 acquisition in progress. QRF calibration is demonstration-grade ($n_{cal} = 70$, ±3-4 pp sampling wobble). None of these caveats touches the protocol, whose steps consume only pre-modelling information. A final boundary is geographic and crop-system scope: every empirical result here comes from Chinese winter-wheat systems, and the protocol itself, while domain-agnostic by construction, has yet to be exercised on other crops and continents; we encourage such transfer tests and release all tooling to enable them.

11. Conclusions

For model-derived yield targets, apparent skill is manufactured by target construction, validation design, and representation limits-not by environmental information, which our own framework confirms to be large ($R^2 = 0.69$) wherever gradients are wide enough to detect. We release a six-step pre-mapping audit protocol: (i) declare target provenance and training fit; (ii) estimate the variogram range from the target field and report domain size in its own range units-checkerboard families are structurally inert near or below one range and across small domains; (iii) screen PC1/elevation before celebrating multi-variable importance; (iv) report interval width against the decision it must support; (v) benchmark the sampling-geometry signal ceiling (domain × environmental variance × n × model complexity) before interpreting a null; and (vi) audit inter-target agreement, with official statistics as arbiter where they exist. Every step costs minutes, requires no new data, and would have caught every trap documented here before publication.

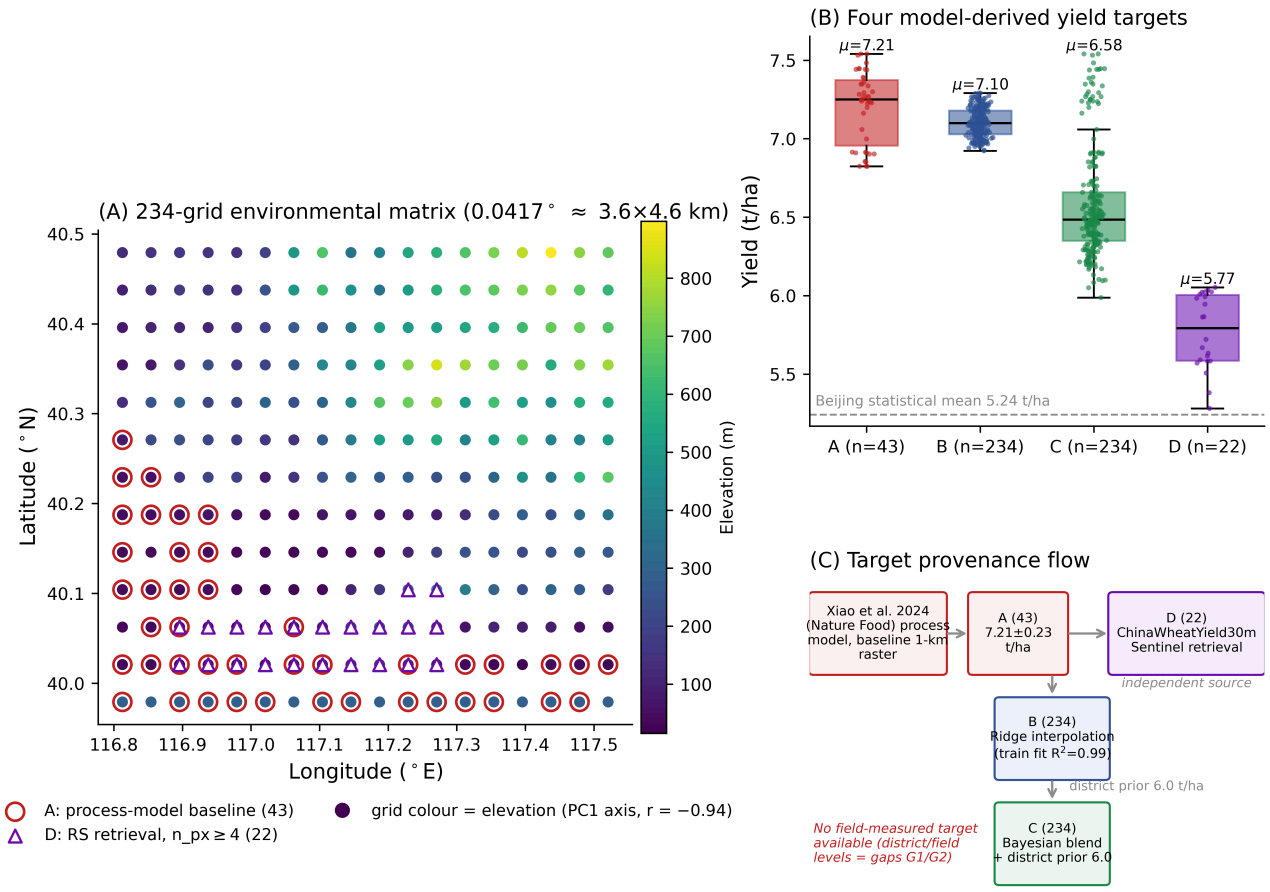


Figure 1. Study area and target provenance. (a) 234-grid environmental matrix coloured by elevation (the PC1 axis), with process-model baseline grids (43) outlined and satellite-retrieval grids (22) marked; (b) distributions of the four model-derived targets against the Beijing statistical anchor; (c) provenance flow from published products to targets A-D, with the ground-truth gap declaration.

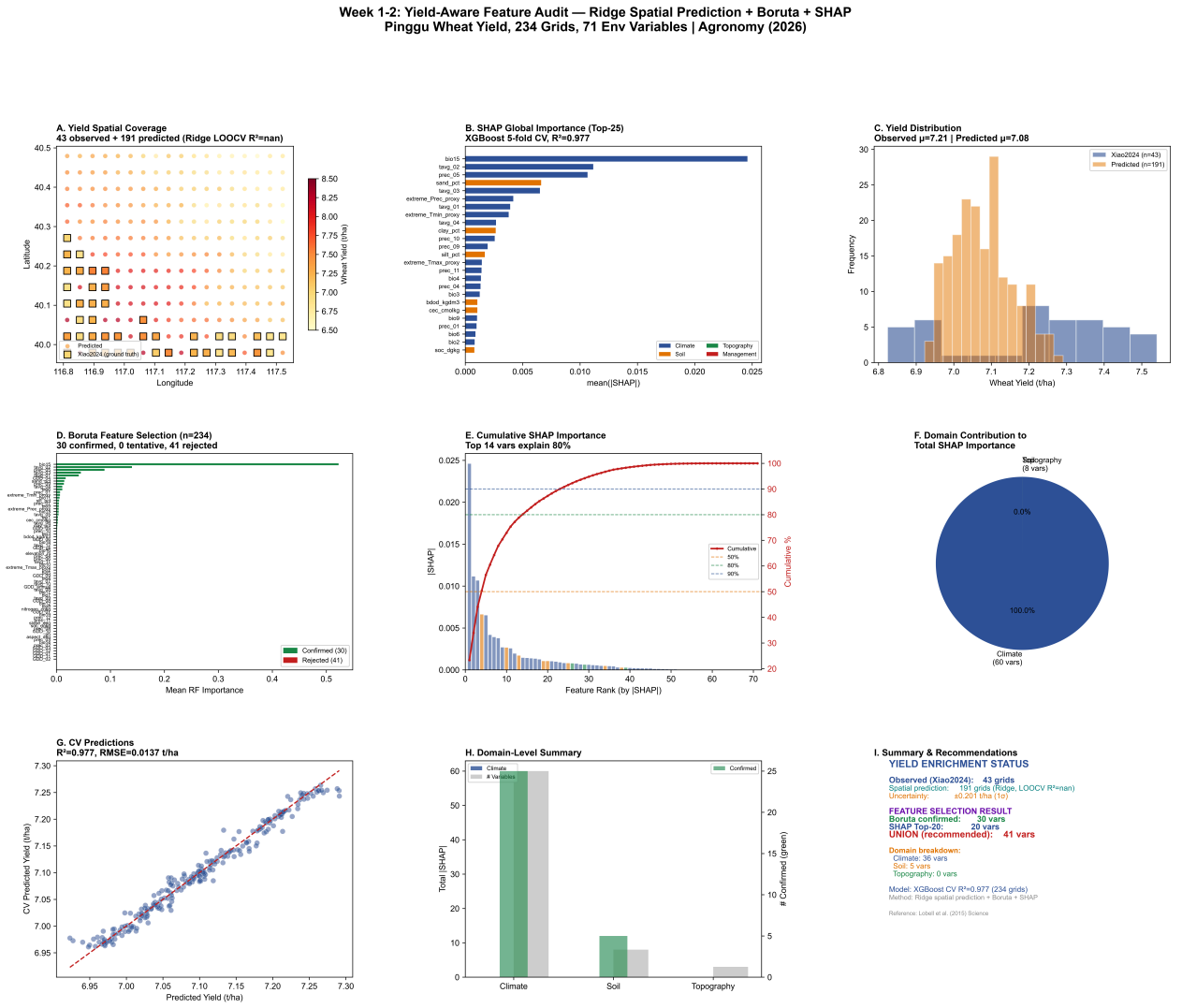


Figure 2. Feature-audit dashboard. Distributions, missingness, and pairwise-correlation structure of the 24 modelling variables across the 234 grids; variance-inflation factors with the screening threshold; Boruta classification (confirmed/tentative/rejected) over 50 iterations with five shadow attributes. The audit that later isolated an SRTM extraction defect in an early data build (18 grids with missing terrain values, concentrated in the warm south-west) is documented in the repository history.

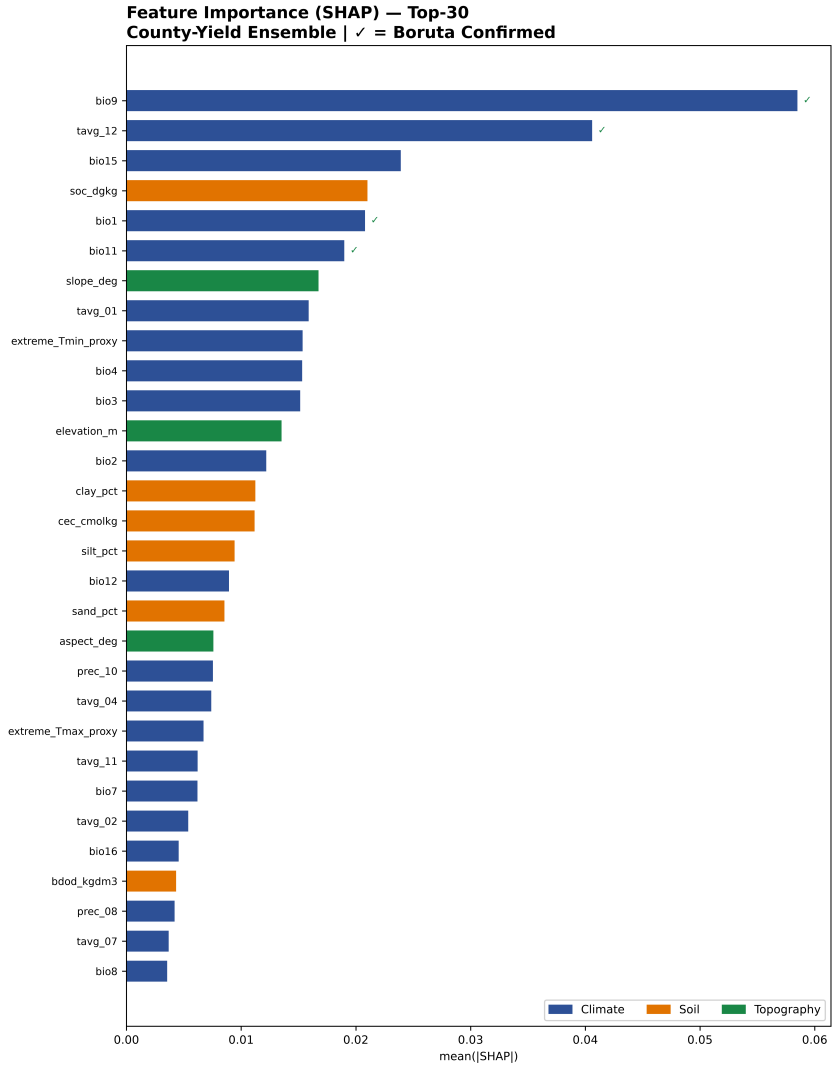


Figure 3. SHAP importance, top 30 variables. XGBoost Shapley values computed on training folds; points coloured by variable domain (climate, soil, topography); check-marks denote Boruta-confirmed variables. The leading variables (bio9, December mean temperature, bio11) are winter-temperature expressions of the single elevation axis identified in Fig 9 and are rank-unstable within that collinear block.

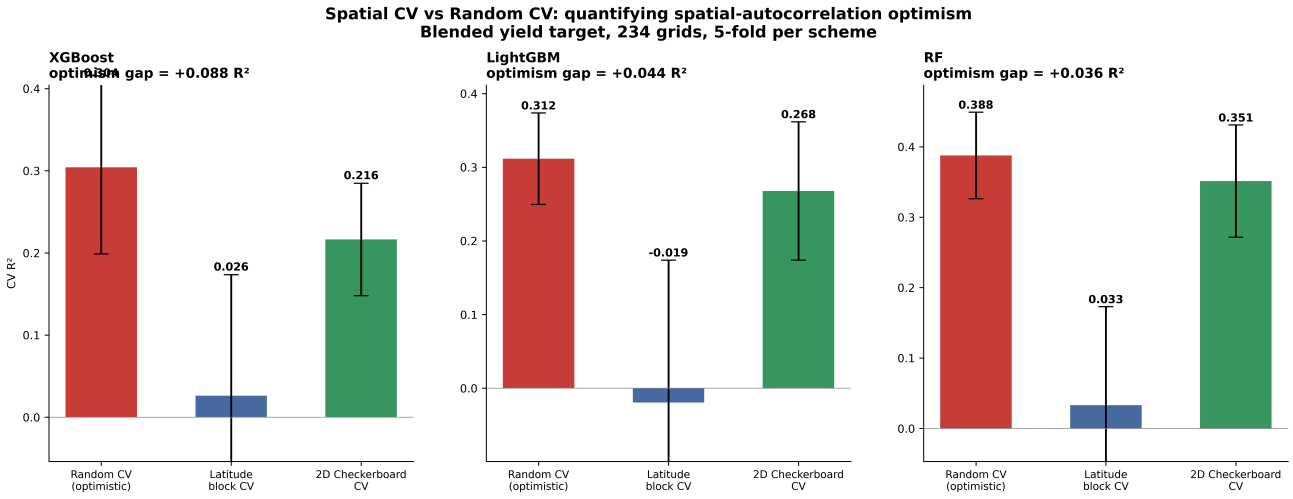


Figure 4. Validation-design spectrum and dose-response. (a) five-scheme spectrum; (b) optimism gap versus block size in target-field range units; (c) the degeneration mechanism: between-block nearest-neighbour distance pinned at 2.3-2.4 km for all checkerboard sizes.

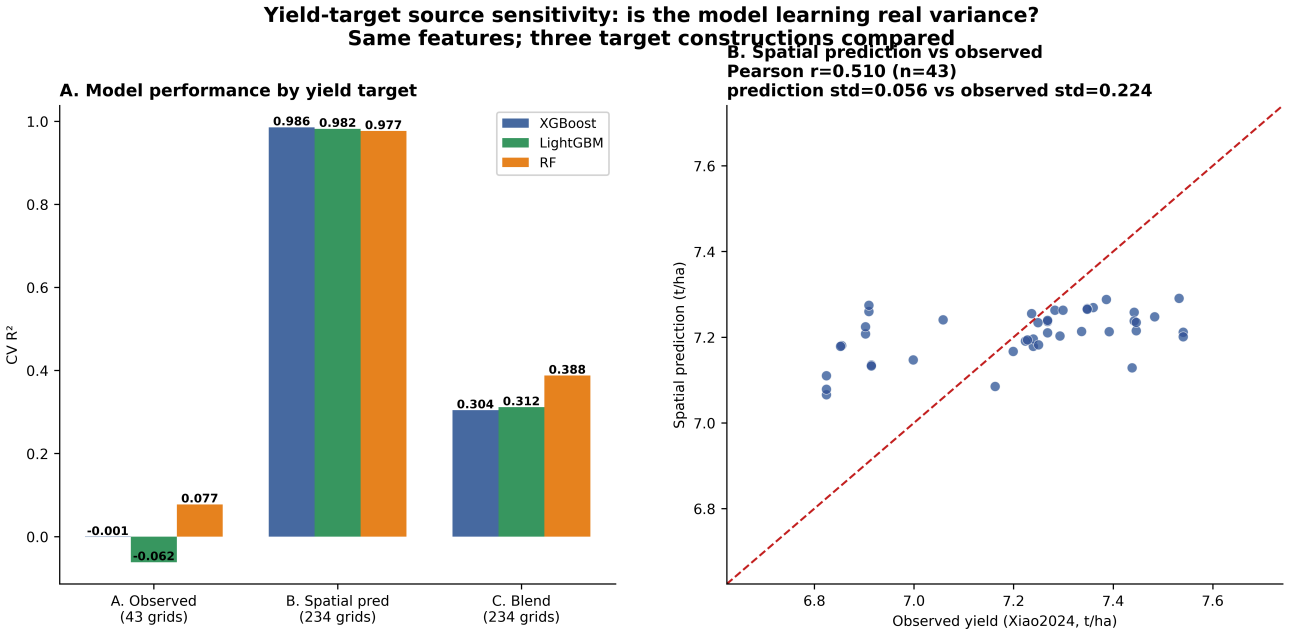


Figure 5. Target-source sensitivity. Pooled cross-validated R2 of the same environmental models against all four targets: A (process-model baseline, n = 43, R2 near zero under every scheme), B (interpolation of A, R2 = 0.98 versus a leave-one-out interpolation bound of 0.20-0.27), C (Bayesian blend, intermediate), and D (satellite retrieval, n = 22, fold-level and pooled statistics diverging exactly as small-block variance predicts).

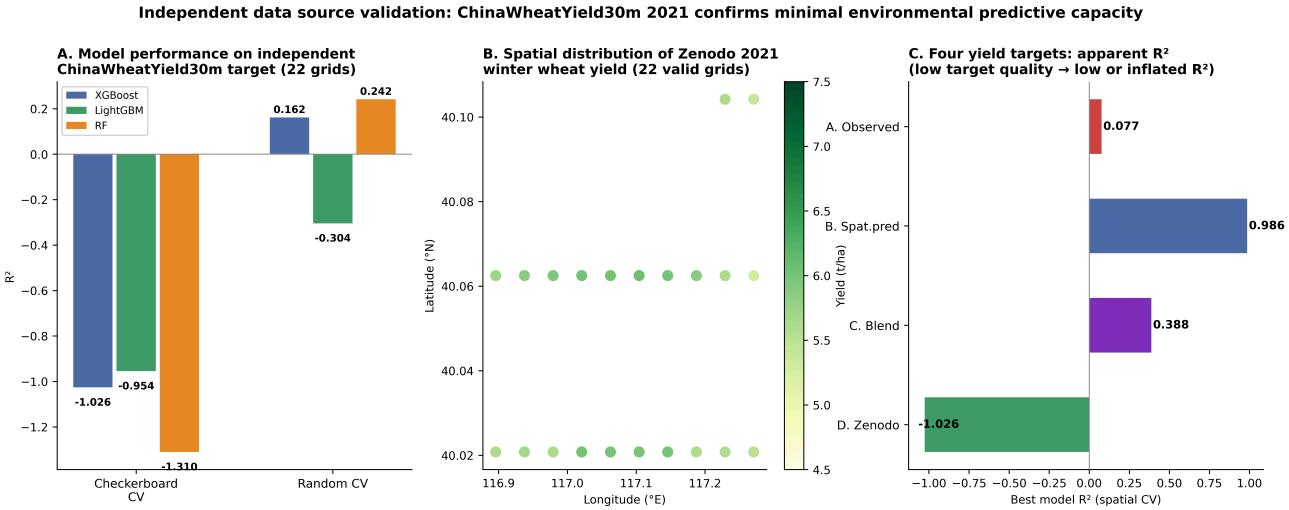


Figure 6. Independent-source validation. Cross-validation of environmental models against the satellite-retrieval target D under random and checkerboard schemes; checkerboard fold-level R2 is negative for all models while pooled statistics remain ambiguous at n = 22.

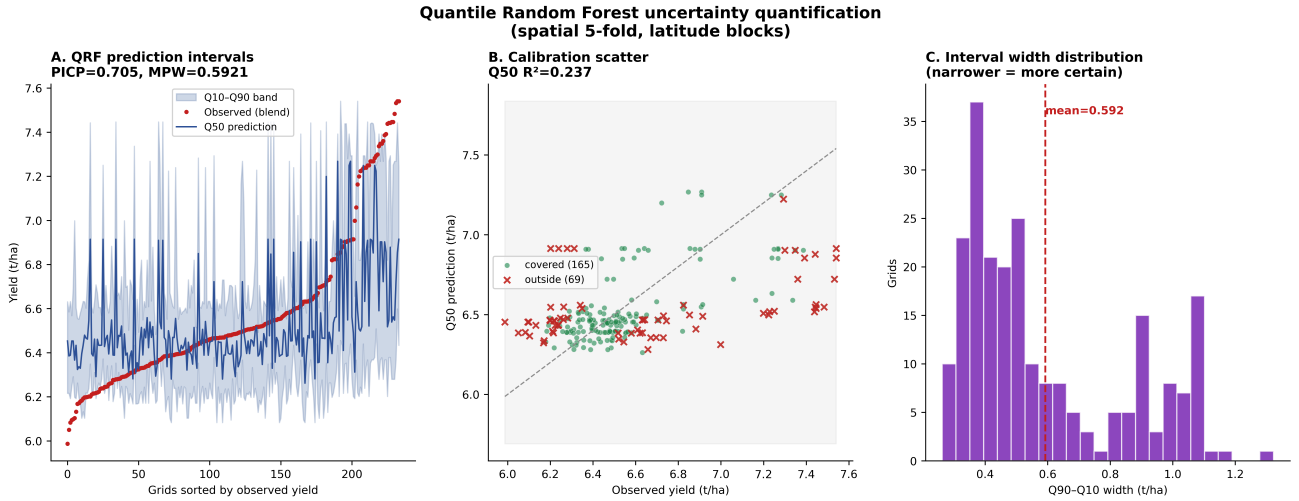


Figure 7. QRF uncertainty and conformal calibration. Prediction-interval bands with observations overlaid (a), calibration coverage versus the nominal 90% (b), and interval-width distribution showing heteroscedastic widening in the high-yield south-west (c); raw PICP 0.705 rises to 0.862 after split-conformal calibration at a width cost of 0.62 to 0.79 t/ha, roughly 2.6 times the target standard deviation.

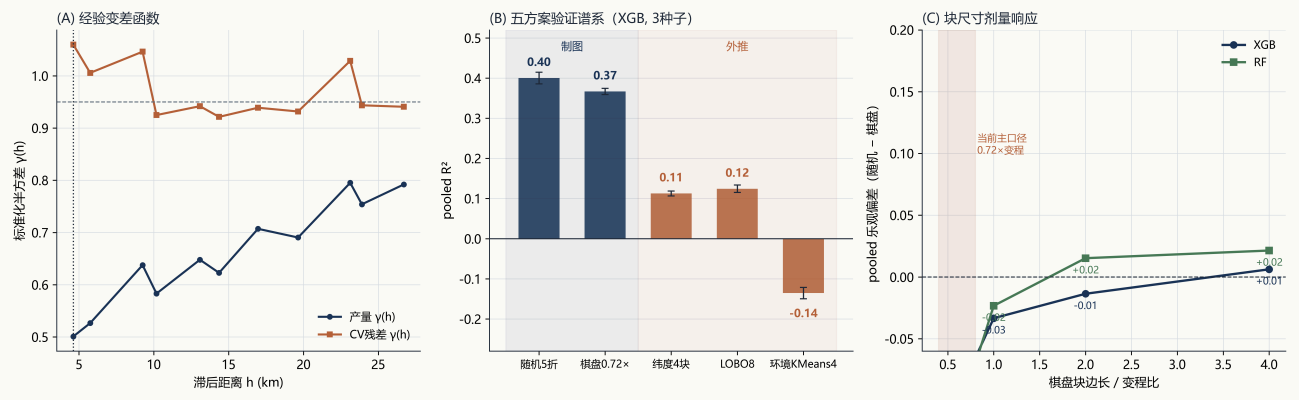


Figure 8. Block-size defence and degeneration mechanism. (a) Empirical variograms of yield and cross-validation residuals (range about 4.6 km); (b) optimism gap versus block size in target-field range units for both domains; (c) the mechanism probe: between-block nearest-neighbour distance remains 2.3-2.4 km for every checkerboard size in the county domain, identifying silent degeneration into random splits.

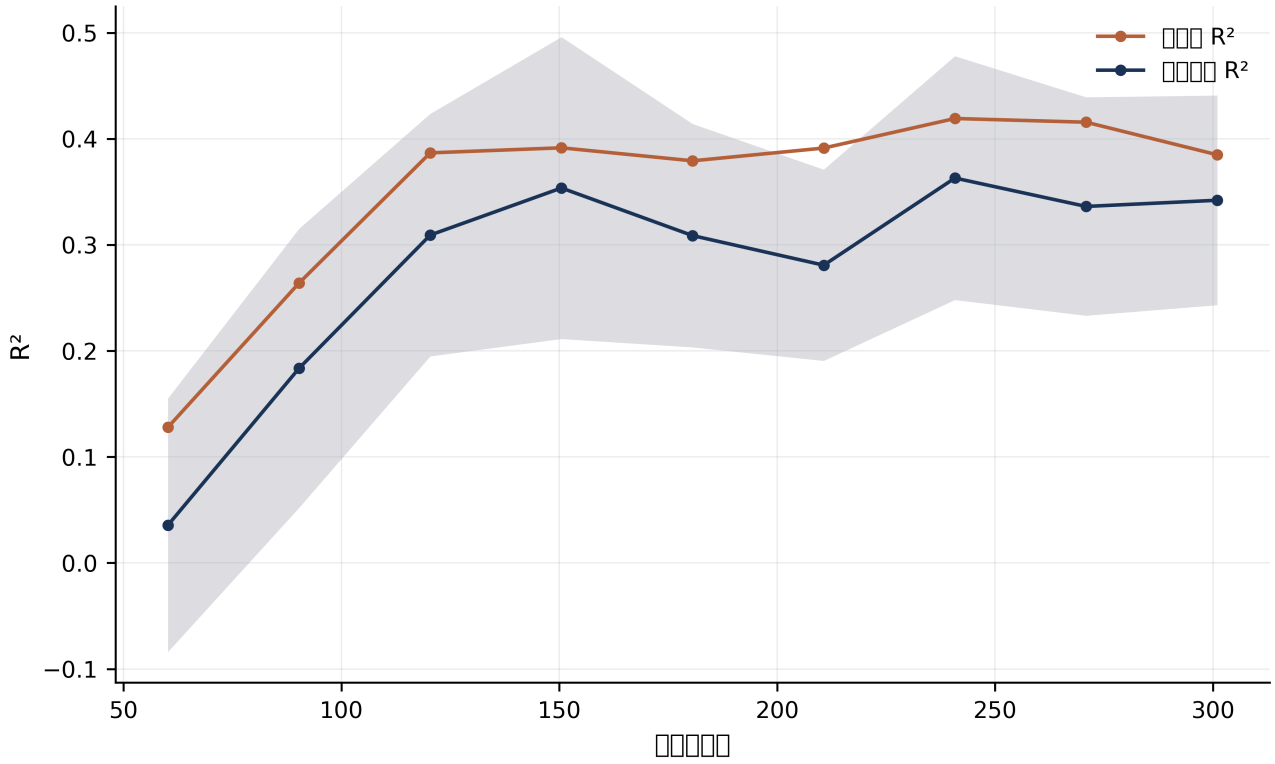


Figure 9. Principal-component diagnostics. Loadings of the 24 variables on PC1 versus elevation (r = -0.94) and per-axis R2 of targets on PC1-PC5, establishing the one-axis structure of the environmental fingerprint.

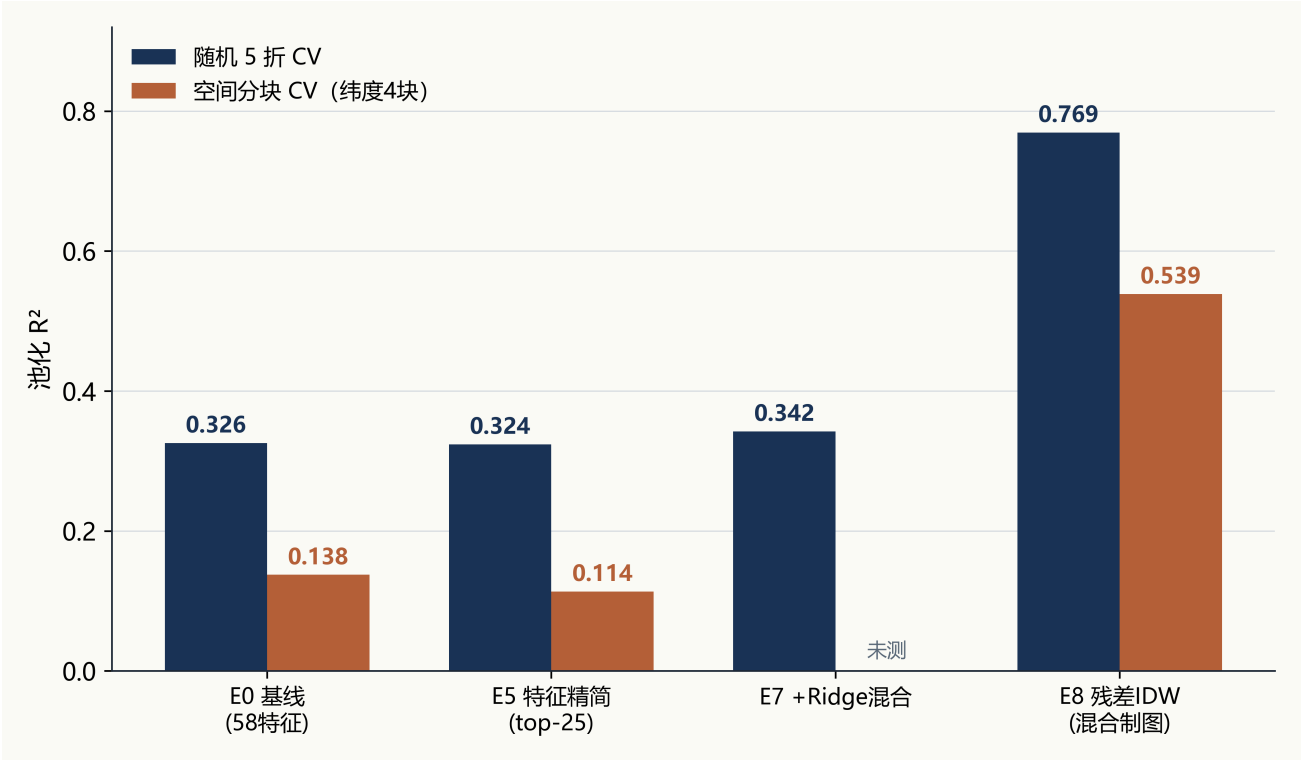


Figure 10. Learning curves of the 250-m hybrid pipeline. Out-of-fold skill versus training-set size for the environmental baseline and the hybrid environmental-residual model.

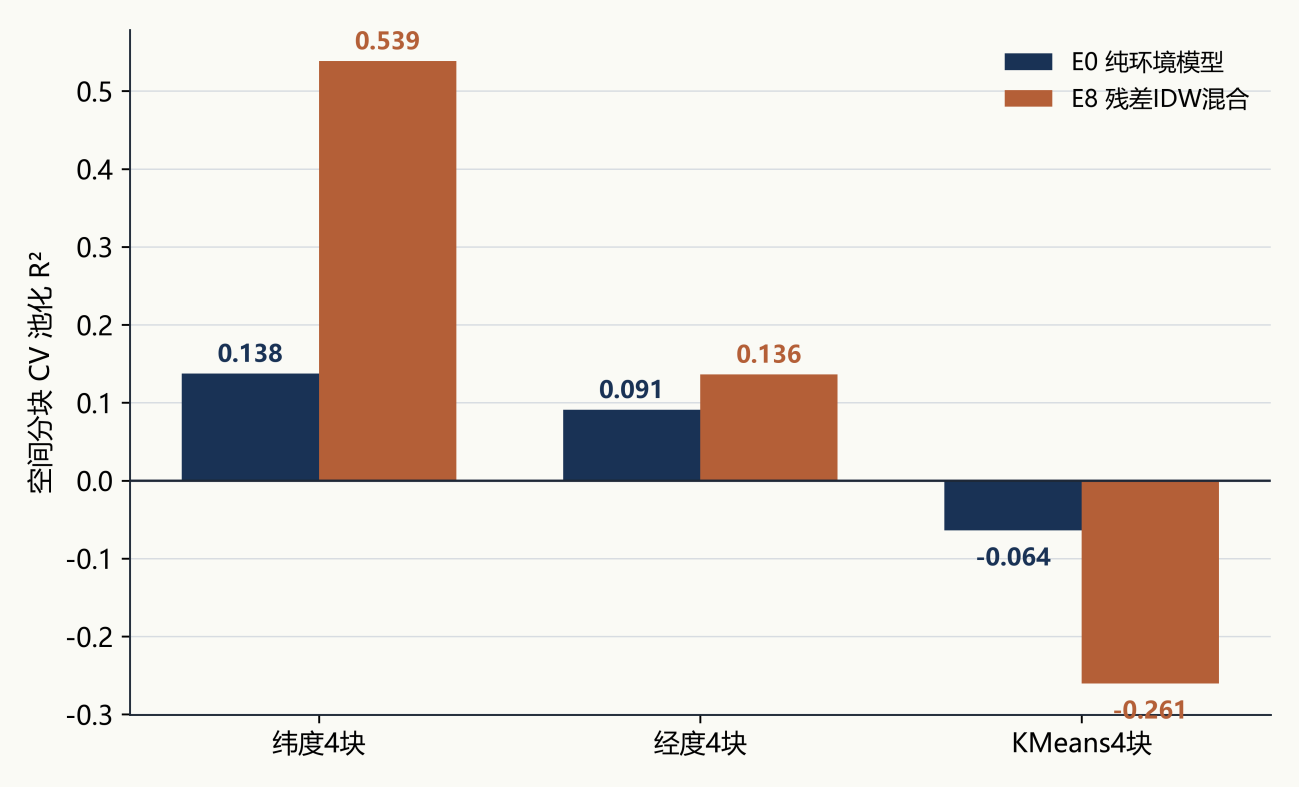


Figure 11. Blocking sensitivity of the 250-m hybrid. Pooled R2 under latitude, longitude, and K-means blocking: hybrid gains vanish when spatial continuity is cut, identifying the gain as interpolation enhancement rather than extrapolation skill.

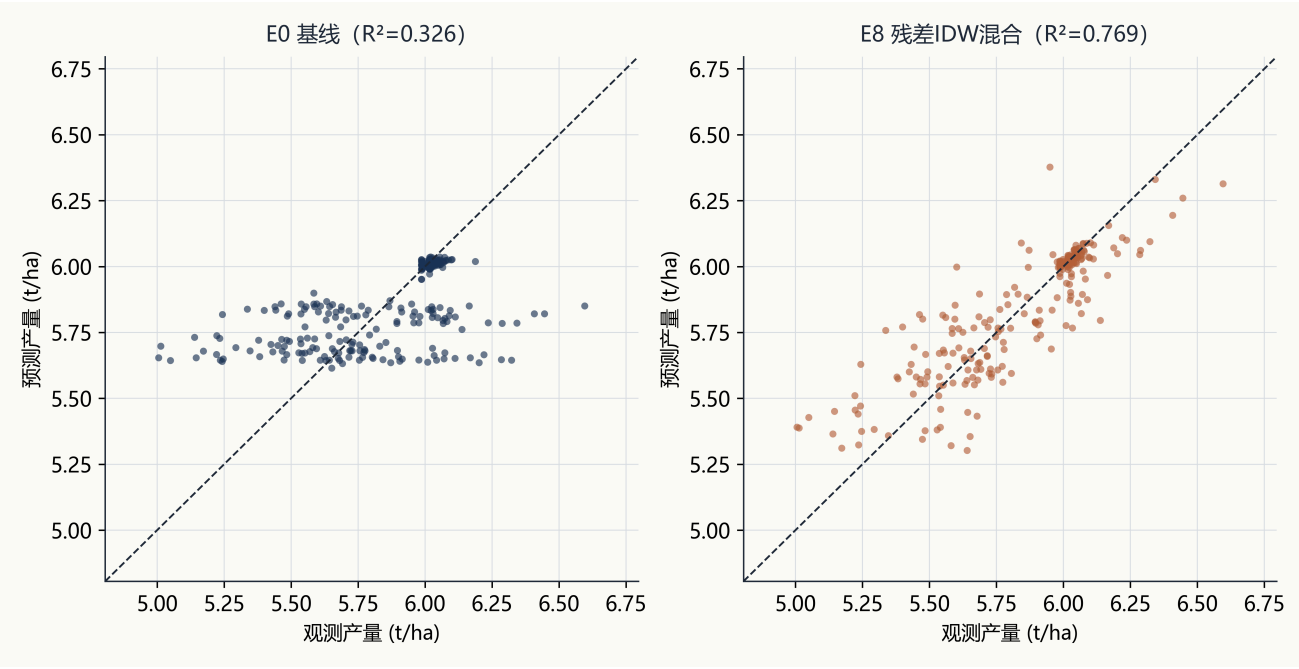


Figure 12. Observed-predicted scatter, 250-m line. Random five-fold out-of-fold predictions for the environmental baseline (R2 = 0.326) versus the hybrid (R2 = 0.769), with the 1:1 line; high-yield underestimation of the baseline is corrected by the residual term.

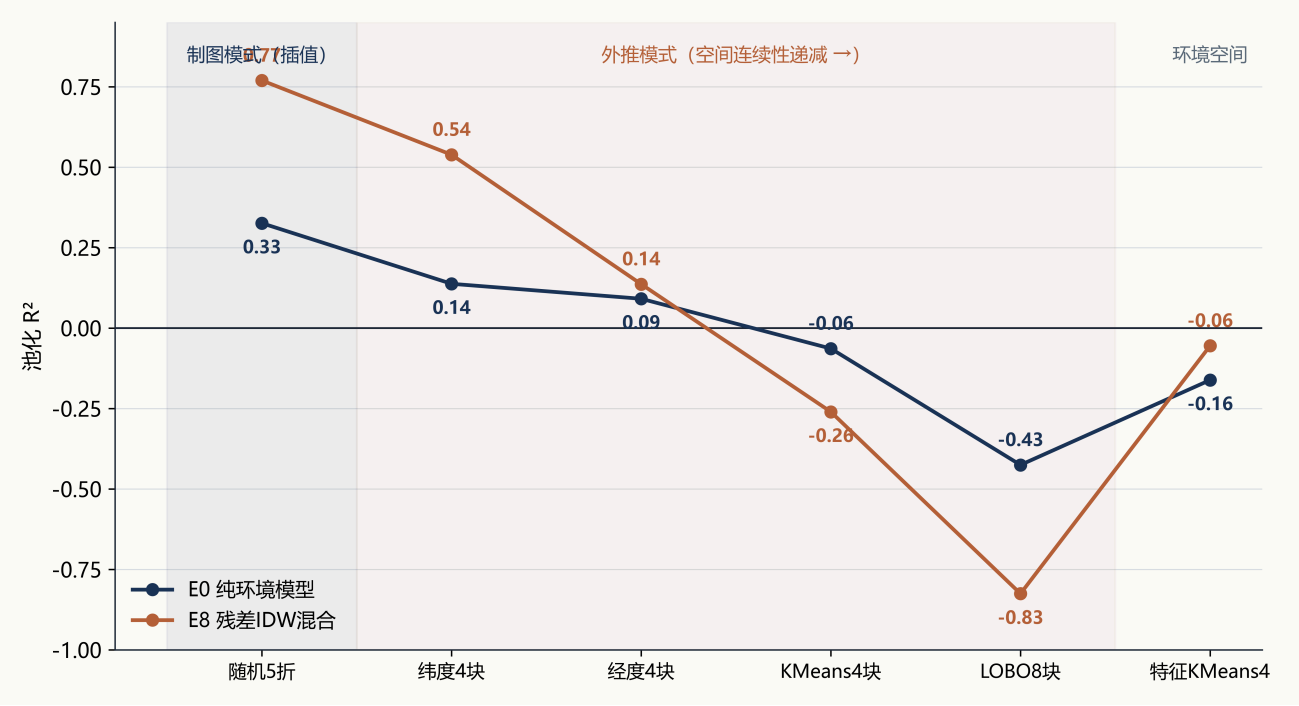


Figure 13. Extrapolation-gradient spectrum. Pooled R2 of baseline and hybrid models across six validation schemes ordered by decreasing spatial connectivity (random, latitude blocks, longitude blocks, K-means, leave-one-block-out, environment-space blocks): monotone decay to -0.83 quantifies the interpolation-versus-extrapolation boundary.

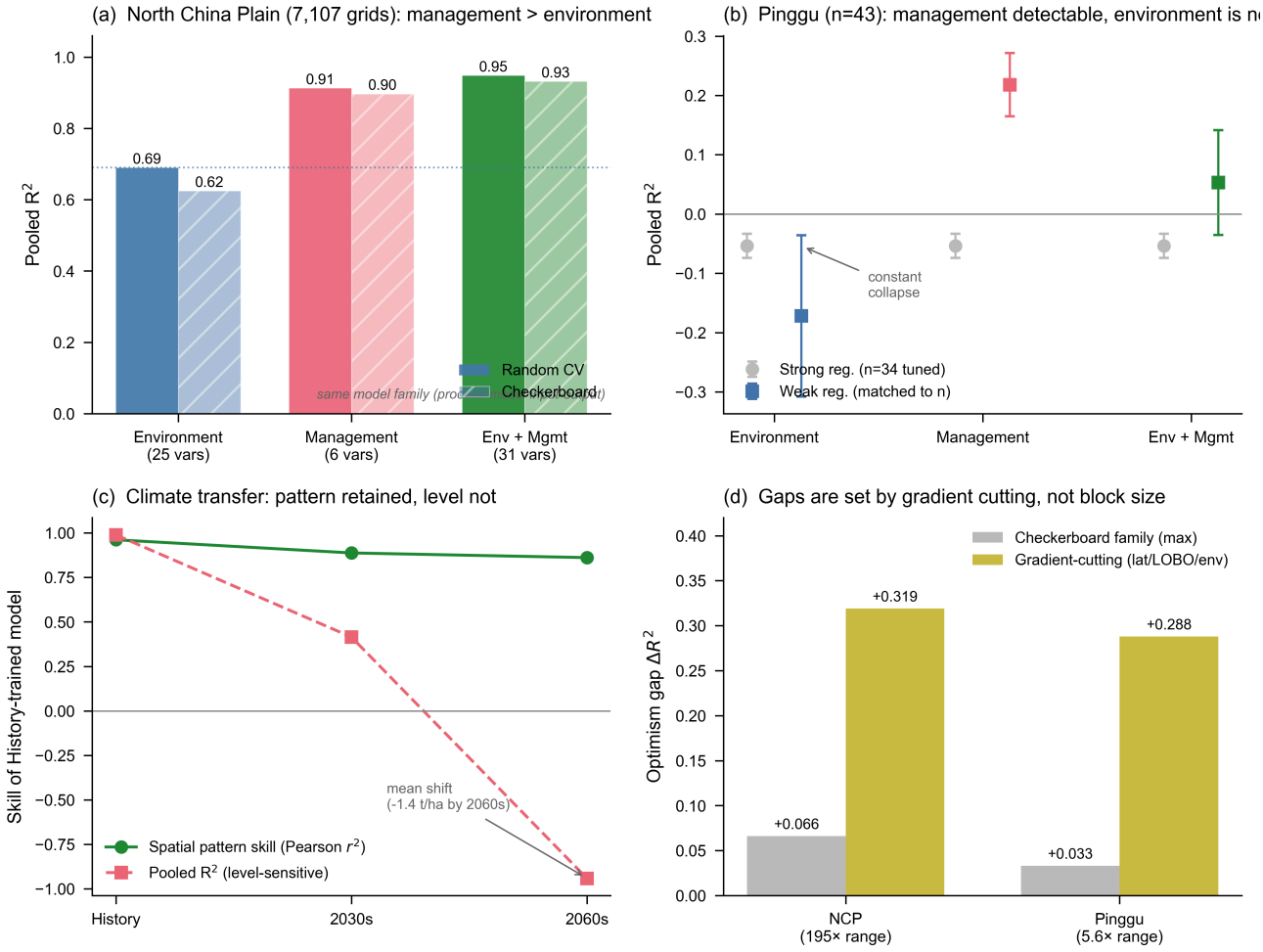


Figure 14. Detectability synthesis. (a) national variance partitioning E/M/E+M; (b) Pinggu n=43 under matched versus over-regularised models; (c) climate transfer (pattern skill vs pooled R2); (d) two-domain optimism gaps (checkerboard family vs gradient cutting).

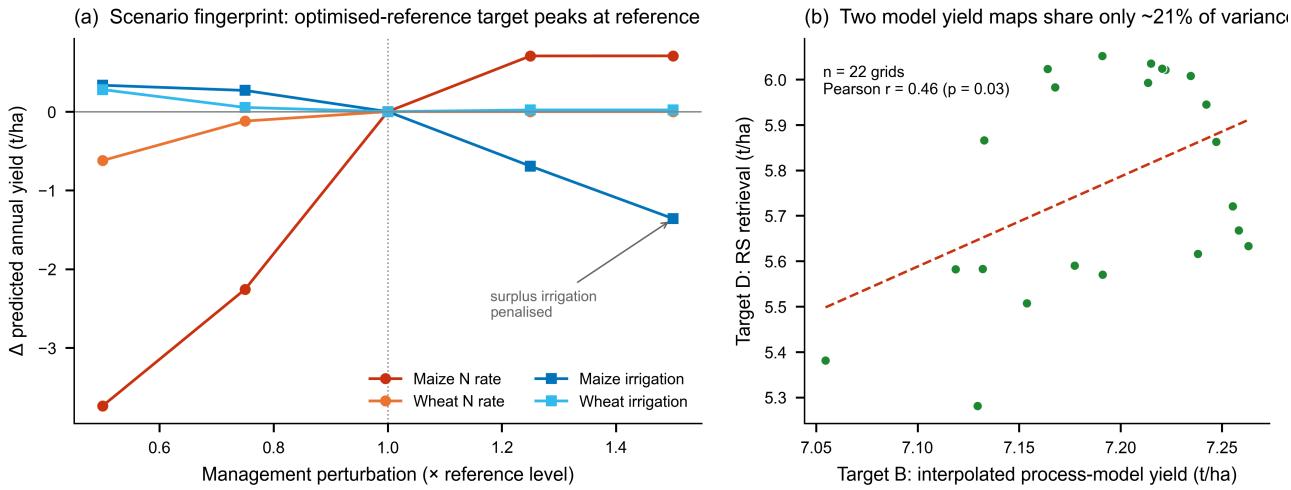


Figure 15. Target-semantics diagnostics. (a) Scenario-response fingerprint: perturbing the management reference surface around its published optimum yields diminishing returns to maize N and a penalty to surplus irrigation, a shape compatible with optimised-reference semantics (and, as the collinearity analysis shows, not uniquely diagnostic of it); (b) inter-target agreement at county scale: the interpolated process-model surface B versus the satellite retrieval D on their 22 common grids (Pearson r = 0.46), with the blend target C showing no spatial structure against D (r = -0.01).

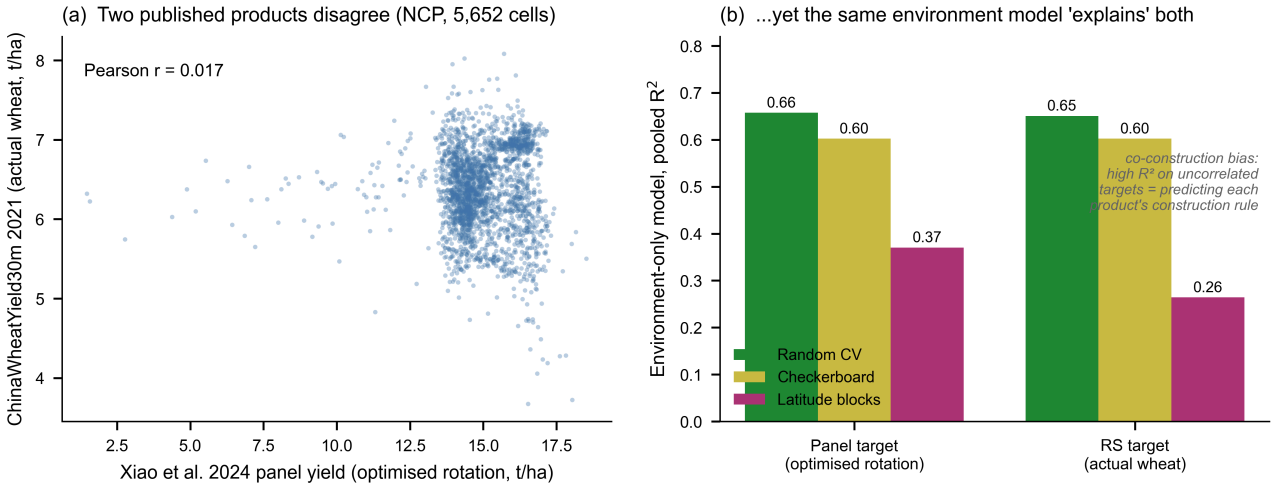


Figure 16. Co-construction bias. (a) two published products disagree (r=0.017, 5,652 cells); (b) the same environmental model "explains" both at R2 near 0.65.

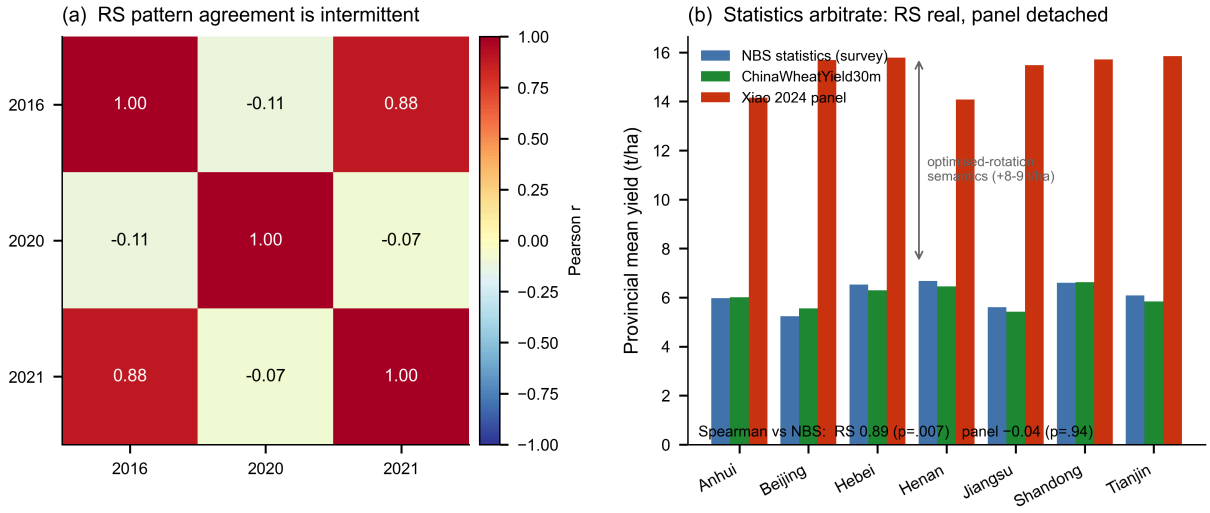


Figure 17. Intermittent inter-annual agreement (pairwise pattern matrix) and provincial statistical arbitration (NBS survey vs products).

Table S1. Six-step pre-mapping audit protocol, self-assessment of this study

Step	Check	This study
i	Target provenance declared; training fit disclosed	All four targets audited; B's training R2=0.99 disclosed; "observed" mislabel corrected
ii	Variogram from target field; domain in own range units; block family justified	Pinggu 4.62 km/5.6x; national 462 km/~2x; checkerboard shown inert in both
iii	PC1/elevation screened before multi-variable importance	PC1=66%, r=-0.94; SHAP read as elevation proxies
iv	Interval width vs decision requirement	Decision-suitability matrix (insurance x / screening ok)
v	Sampling-geometry signal ceiling benchmarked before null interpretation	Window sweep + planted-signal power analysis
vi	Inter-target agreement audited; statistics arbitrate	r=0.017 alarm; NBS rho=0.893 (RS) vs -0.036 (panel)

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Declarations

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Data Availability Statement: All data are public: WorldClim 2.1, SoilGrids 2.0, SRTM (public domain), the Xiao et al. panel (CC BY 4.0, figshare 10.6084/m9.figshare.24471919), ChinaWheatYield30m (CC BY 4.0, Zenodo 10.5281/zenodo.7360753; years 2016-2021), Open-Meteo ERA5-Land (CC BY 4.0), and National Bureau of Statistics provincial yields. File-level SHA-256 checksums are provided in the data manifest; rasters ship via the repository data release; fold assignments, seeds, parameters, and all statistical ledgers are frozen in the code repository, and every number in this paper regenerates from a scripted ledger.

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